

Improving HTR Attention Maps with Register Tokens:

A CTC Vision Transformer for Interpretable Handwritten Text Recognition

— Progress Draft —

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Abstract

Handwritten Text Recognition (HTR) on the IAM benchmark is close to saturated on character error rate (CER): a compact CNN–RNN–CTC baseline reaches 4.5–4.9% CER on the Aachen split. What is not saturated is *attention interpretability*. Self-attention maps from Vision Transformer (ViT) recognizers are noisy and dominated by a small number of high-norm “sink” patch tokens, which limits their usefulness for downstream tasks such as writer identification. This report tests whether the register-token mechanism of Darcet *et al.*, originally proposed for classification ViTs, transfers to a CTC-based HTR system and improves attention quality without hurting recognition. We build a CNN-stem Vision Transformer with a dual BiLSTM+CNN CTC head (ViT-RGTS v2, 9.5M parameters), sweep the register count over $R \in \{0, 2, 4, 8, 16\}$, and compare against the CNN–RNN baseline, a pretrained TorchVision ViT-B/16 fine-tuned with a two-group learning rate, and TrOCR fine-tuned with layer-wise learning-rate decay. In total we run 63 training experiments (runs 143–205, up to 150 epochs each) on IAM and READ2016, with three seeds for a variance floor and 16 synthetic-pretraining runs on $\sim 950\text{K}$ rendered lines.

Two findings organize the report. First, on the recognition axis the register-token count is close to neutral: the test CER varies by only 0.24 pp across $R = 0$ to $R = 16$ on IAM (range 5.74–5.98%, tightening to 5.60–5.70% once the delivered SWA-averaged model is evaluated) and by 0.09 pp on READ2016, both within a seed-noise floor of ± 0.08 pp measured over three seeds. This matches Darcet *et al.*’s own observation for classification ViTs that registers are an interpretability intervention rather than an accuracy intervention. Second, on the attention axis the claim only partially transfers: register tokens visibly absorb the artifact-attention pattern seen in the $R=0$ ViT, and per-character attention becomes sharper and better aligned with the CTC decoding column, but a residual sink pattern intrinsic to CTC self-attention remains and requires an additional CTC-column prior to suppress. The best overall recognizer stays the CNN–RNN baseline with SWA (4.50% CER, 15.30% WER, 7.36M parameters), and the best interpretable recognizer is ViT-RGTS v2 with four registers and SWA (5.70% CER, 18.70% WER, 9.48M parameters). Two ablations collapse to $\sim 73\%$ CER when the CNN stem is removed or when a legacy row-major patch layout is used, both isolating the CTC-monotonic-alignment constraint as the single load-bearing design decision for a Transformer HTR encoder. All models, configurations, and 63 experiment logs are released with the report.

Index Terms— *Handwritten Text Recognition, Vision Transformer, Register Tokens, Attention Interpretability, CTC, IAM, READ2016, Writer Identification.*

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1 Introduction

Handwritten Text Recognition (HTR) has matured to the point where character error rates (CER) on the standard IAM benchmark [8] sit in the low single digits: the CNN–RNN–CTC baseline of Retsinas *et al.* [12] reaches $\sim 4.6\%$ CER without any pretraining or synthetic data. With recognition largely solved on the widely used benchmarks, the open question in HTR shifts from *whether* the model reads the line correctly to *what the model looks at when it reads*, and whether those internal signals are clean enough to reuse for other document-analysis tasks such as writer identification, style analysis, and palaeographic dating that traditionally required hand-designed feature pipelines [13].

This report follows the second question. The specific brief from the supervisor, restated verbatim from the project’s opening emails, was: “*make HTR explainable as much as possible for potential downstream tasks such as writer identification — attention maps should be fine-grained and meaningful, and register tokens from the Vision Transformers Need Registers paper should be integrated to test their effect on attention quality.*” The rest of Section 1 unpacks that brief into a background, a research gap, four research questions, and a list of contributions.

1.1 Background and Motivation

Modern HTR pipelines fall into two structural families. The first is the CNN–BiLSTM–CTC family, canonically represented by Retsinas *et al.* [12]: a convolutional stem collapses the input line image to a one-dimensional sequence, a recurrent head models sequential dependencies, and Connectionist Temporal Classification (CTC) [3] handles the length mismatch between output characters and input positions. This family is small, fast, and works well on limited training data because CNN inductive biases (locality, translation equivariance) match the physical structure of ink strokes. The second family is Transformer-based: either encoder-only with CTC (as in HTR-VT [7]) or encoder–decoder with cross-attention (as in TrOCR [6]). These trade CNN inductive bias for a global self-attention mechanism that, in principle, exposes exactly the kind of internal signal — *where on the image is each output character being read from* — that downstream applications need.

In practice, this internal signal is not clean. Darcet *et al.* [1] showed that Vision Transformers trained on standard classification tasks develop a small number of anomalously high-norm patch tokens that behave as *attention sinks*: they absorb a disproportionate share of the attention mass across layers, produce visible artifacts in attention maps, and hurt any downstream task that reads those maps. Their proposed fix is minimally invasive — prepend a small number of learnable *register tokens* to the patch sequence, and the sinks migrate onto the registers, leaving the patches free to produce interpretable attention. On classification ViTs the fix is clean: it improves downstream detection and segmentation while leaving classification accuracy essentially unchanged.

Whether the same fix works for a CTC-based HTR system is not obvious, and this is the empirical question the report addresses. HTR self-attention differs from classification self-attention in two structural ways. First, HTR has no dedicated CLS query: the queries at inference are the patch tokens themselves, so the “sink” phenomenon — if it appears at all — is queried by the same population of tokens that produce it, which is a different setting than the CLS-vs-patch dichotomy Darcet *et al.* analyse. Second, the CTC loss is agnostic to attention structure: it only cares whether some monotonic alignment of the character labels through the token sequence has high probability, and does not directly penalize an attention map that “looks bad” as long as the decoding is correct. Both differences are reasons to expect that porting registers from classification ViTs to CTC HTR is not free.

1.2 Problem Statement

The project has one deliverable and one measurable question.

- **Deliverable.** A modular, reproducible HTR training and evaluation pipeline that supports CNN–RNN, ViT (from scratch), and pretrained-backbone models within a single interface, with attention-extraction hooks aligned to the visualization style of Fig. 5 in *Beyond Memorization* [11].
- **Measurable question.** Does adding $R \in \{0, 2, 4, 8, 16\}$ register tokens to a CTC ViT HTR system (a) change the character/word error rate on the IAM Aachen split and the READ2016 test set, and (b) change the quality of the attention maps produced by the encoder, judged both qualitatively (per-character maps in the style of [11]) and quantitatively (attention concentration on the CTC decoding column of each character)?

1.3 Research Gap

The registers-for-attention idea is well established for classification ViTs and for dense-prediction tasks derived from them (segmentation, detection) [1]. It has not been systematically evaluated for HTR, which uses a sequence-labelling loss (CTC) rather than a per-image class posterior. Two consequences follow. First, no prior work quantifies the recognition-accuracy cost or benefit of registers in the CTC HTR setting. Second, no prior work characterizes what registers actually *do* to attention in that setting — whether the sink pattern documented for classification ViTs appears at all in CTC ViTs, and if so whether registers absorb it as cleanly as they do for classification.

The nearest concurrent work, HTR-VT [7], is a CTC ViT for HTR and validates that ViT encoders can match CNN baselines on modern benchmarks, but it does not use registers and does not analyse attention quality. TrOCR [6] exposes clean cross-attention maps because it is encoder–decoder, but that architectural choice is orthogonal to the registers question and comes with a $10\times$ larger parameter budget. The literature therefore leaves a gap that the CTC ViT + registers combination fills naturally: a small, fast HTR encoder whose self-attention is expected to be interpretable by construction.

1.4 Research Questions

The report answers four questions in order:

1. **RQ1 (Recognition)**. How does the register count R affect CER and WER on IAM and READ2016, and are any observed differences larger than the seed-noise floor?
2. **RQ2 (Interpretability)**. Does increasing R visibly reduce the attention-sink pattern in the CTC ViT encoder, and do per-character attention maps become better localized on the character they decode?
3. **RQ3 (Architecture)**. Which architectural choices are load-bearing for a CTC ViT HTR encoder (CNN stem, dual-head, RNN depth, patch layout)? Which are cosmetic?
4. **RQ4 (Comparators)**. How does the CTC ViT with registers compare, at equal training budget, against (i) a compact CNN–RNN baseline, (ii) a large ImageNet-pretrained ViT-B/16 fine-tuned on IAM, and (iii) a large in-domain-pretrained TrOCR-base fine-tuned on IAM?

1.5 Contributions

This report contributes:

1. A **single-framework HTR pipeline** (HTR-Pipeline) that hosts four architecture families — CNN–RNN, ViT-RGTS (with configurable R), pretrained TorchVision ViT-B/16, and pretrained TrOCR — behind a shared training, evaluation, and attention-extraction interface, and an additional HTR-VT lane for the READ2016 external comparison.
2. A **systematic register-token sweep** on IAM ($R \in \{0, 2, 4, 8, 16\}$, with and without stochastic weight averaging) that quantifies the CER-neutrality of registers for CTC HTR: the no-SWA CER range across R is 5.74–5.98% (0.24 pp), tightening to 5.60–5.70% (0.10 pp) once the delivered SWA-averaged model is evaluated, and the seed-noise floor is ± 0.08 pp over three seeds, all established from 150-epoch runs.
3. A **cross-dataset replication** on READ2016 that confirms the same near-flat CER-versus- R pattern (0.09 pp range, 4.73–4.82% CER), separating the finding from any IAM-specific artifact.
4. An **attention-quality analysis** showing that register tokens do reduce the sink pattern qualitatively, that per-character CTC-column attention becomes measurably sharper with $R \geq 4$, and that a residual sink pattern remains — traced to the fact that CTC-ViT patch tokens are simultaneously queries and keys of the sink attention, unlike the classification-ViT setting in which the CLS query and the patch sinks are structurally distinct.
5. An **architecture ablation** that identifies the CNN stem and the strictly-left-to-right patch order it produces as the single non-negotiable design decision for a CTC ViT HTR encoder: removing the CNN stem (row-major 16×16 patches) collapses CER from 5.96% to 73.36%, and the legacy v1 architecture with the same row-major layout independently collapses to 73.65% CER — two convergent failures on the same underlying cause.
6. A **fine-tuning study** for the two pretrained comparators demonstrating that a two-group differential learning rate is required for ImageNet-pretrained ViT-B/16 (15.09% CER with two-group LR vs. 73.14% with naive uniform LR) and that layer-wise learning-rate decay is required for TrOCR (12.10% CER with LLRD+SWA), while gradual unfreezing fails for both.

7. An **honest domain-gap measurement** for synthetic pretraining: models trained purely on $\sim 950\text{K}$ synthetic lines reach $\sim 0.5\text{--}0.8\%$ CER on synthetic-validation but $30\text{--}46\%$ CER on real IAM test, quantifying the render-to-real gap for the specific WordStylist-style synthetic corpus [10] used here.
8. **Full reproducibility artifacts:** all 63 training-run configurations, best-validation checkpoints, per-run `results.csv/training.log` pairs, attention weights for the register sweep runs, and the LaTeX result tables (`unified_results.tex`, `register_sweep.tex`, `ablations.tex`, `architecture_comparison.tex`) shipped alongside the report.

The rest of the report is organized as follows. Section 2 surveys the HTR and register-token literature and positions this work against it. Section 3 describes the datasets, the ViT-RGTS v2 architecture, the register mechanism, and the training procedure. Section 4 lays out the 63-run experiment matrix and the evaluation protocol. Section 5 reports the recognition, ablation, and attention-quality results, and includes a dedicated critical section on whether the Darcet *et al.* claim holds in the CTC HTR setting. Section 6 interprets the findings, and Section 7 closes with directions for extending the pipeline toward writer identification.

2 Related Work

Three research threads converge in this report: Handwritten Text Recognition (HTR) with sequence losses, Vision Transformers and the register-token intervention that improves their attention quality, and attention-based interpretability for downstream document tasks. This section covers each thread and ends with an explicit positioning of the CTC-ViT-with-registers combination against the closest published work.

2.1 Handwritten Text Recognition

Modern offline HTR pipelines are dominated by two families. The first is the CNN-BiLSTM-CTC family, in which a convolutional stem extracts local stroke features, a bidirectional recurrent head models sequence dependencies, and Connectionist Temporal Classification (CTC) [3] resolves the length mismatch between the fixed output grid and the variable-length character sequence. The reference implementation for this family is the “Best Practices” baseline of Retsinas *et al.* [12], which combines three modifications — aspect-ratio-preserving input, column-wise max-pooling to collapse the height axis into a 1D token sequence, and an auxiliary CTC short-cut branch — to reach $\sim 4.6\%$ CER on IAM with a compact CNN-LSTM model. This baseline directly underpins the CNN-RNN branch of our pipeline and defines the “low-CER” bar against which the ViT branch is measured. A broader survey of the HTR field is given by Memon *et al.* [9], who catalog the shift from HMM-based recognizers to the current CNN-recurrent and Transformer-based systems, along with the standard dataset landscape (IAM, RIMES, READ) and evaluation protocols (CER, WER).

The second family replaces the CNN encoder with a Vision Transformer (ViT). HTR-VT [7] keeps a CNN stem for the initial 2D-to-1D reduction but delegates sequence encoding to a plain ViT encoder with a small span-masking regularizer, again supervised by CTC; it demonstrates that ViT encoders can match or exceed CNN-LSTM performance on IAM, READ2016, and other benchmarks at a comparable parameter budget. TrOCR [6], in contrast, replaces the entire pipeline with a pretrained ViT encoder-decoder trained on 684M synthetic and real handwriting lines. TrOCR reaches very low CER but at a $\sim 90\times$ larger parameter count than the Retsinas baseline, and its attention structure — driven by autoregressive cross-attention from character queries to image keys — is architecturally distinct from CTC self-attention. Both HTR-VT and TrOCR are used as external comparators in Section 5; neither integrates register tokens, and neither analyses attention quality under a register-count sweep.

2.2 Vision Transformers and Register Tokens

The Transformer architecture [14] and its Vision Transformer variant [2] replace convolutional feature extraction with a stack of multi-head self-attention blocks that operate on a flattened patch sequence. In a canonical classification ViT, a special CLS token is prepended to the patch sequence, self-attention aggregates image content into it across layers, and a linear head decodes the final CLS representation into a class posterior. Attention maps from the last few layers of such a model are commonly inspected for interpretability — either as a saliency proxy or as the primary output of dense-prediction methods derived from ViTs.

Darcet *et al.* [1] identified a specific failure mode of this interpretability. In sufficiently well-trained supervised and self-supervised ViTs, a small number of patch tokens develop anomalously high norms and disproportionate attention mass; these tokens act as *attention sinks*, absorbing global information the model would otherwise have no place to store. The visible symptom is a set of bright, spatially arbitrary artifacts in the attention maps that do not correspond to any semantically meaningful part of the image. Their fix is to prepend R additional learnable tokens, called *register tokens*, to the patch sequence. The sinks migrate onto the registers during training, leaving the patch tokens with cleaner, more localized attention. On classification ViTs, adding $R \in \{4, 8, 16\}$ registers leaves top-1 accuracy essentially unchanged (deltas within noise) while measurably improving downstream detection and segmentation from the same features. The register mechanism does not require architectural surgery, does not add a new loss term, and adds only $R \cdot D$ parameters to a model of embedding width D .

The claim by Darcet *et al.* is stated for classification ViTs and dense-prediction models derived from them. It has not been evaluated for sequence-labelling ViTs supervised by CTC. Two structural differences separate the two settings. First, a CTC ViT has no CLS query — the patch tokens are simultaneously the queries and the keys of the attention operation that produces the output logits, so any sink behaviour is produced by the same token population that reads it. Second, the CTC loss is agnostic to attention structure and does not directly penalize a map that “looks” bad if the decoded character sequence is correct. Whether registers still absorb the sinks in this setting, and whether that absorption translates into cleaner per-character attention maps, is the empirical question of this report.

2.3 Attention-Based Interpretability and Downstream Writer Analysis

The interest in clean HTR attention maps is driven by two concrete downstream applications: attention-based character localization for synthetic-handwriting generation, and character-indexed feature aggregation for writer identification.

The first application is developed by Nikolaidou *et al.* in *Beyond Memorization* [11], a training-free style-mixing method for handwritten text generation built on top of a pretrained diffusion model. Their Figure 5 shows per-character attention maps localized on individual glyphs in the rendered output — the visualization style that Vincent explicitly asked this project to reproduce in an HTR-recognition setting. Two caveats deserve statement now, so that Section 5 can be read against them. First, the *Beyond Memorization* maps are cross-attention maps from a diffusion U-Net, in which explicit text-token queries attend to image-region keys. Our HTR self-attention setting has no equivalent text-token query at inference: the queries are the patch tokens themselves, and character positions are recovered from the CTC decoding column rather than from a text-side query slot. Second, the Nikolaidou *et al.* system operates on word-crop images (roughly 256×64) whereas HTR line images are $\sim 1024 \times 128$ — an aspect ratio that produces thin horizontal attention grids rather than the near-square grids the *Beyond Memorization* visualization was designed for. Section 5 treats these differences as a genuine finding of the project, not as a limitation to hide.

An earlier precursor of the same line of work is WordStylist [10], which introduced the writer-embedding-injection idea used by *Beyond Memorization*. WordStylist is not a direct comparator

Table 1: Positioning against the closest published work.

System	Loss	Encoder	Registers	Attention analysis	HTR data
Retsinas <i>et al.</i> [12]	CTC	CNN-BiLSTM	—	—	IAM, RIMES
HTR-VT [7]	CTC	CNN+ViT	—	no register sweep	IAM, READ2016, ...
TrOCR [6]	Cross-entropy	Pretrained ViT+dec.	—	cross-attention viz	684M synth+real
Darcet <i>et al.</i> [1]	Classification / SSL	ViT	$R \in \{4, 8, 16\}$	attention/sink analysis	ImageNet
This report	CTC	CNN+ViT	$R \in \{0, 2, 4, 8, 16\}$	register sweep, per-char maps	IAM, READ2016 (+ synth.)

here, but its rendered-line pipeline is close in spirit to the synthetic-handwriting corpus of $\sim 950K$ lines used in the pretraining lane of Section 5.

The second downstream application is writer identification via character-indexed feature aggregation. Souibgui *et al.* [13] introduce Vectors of Locally Aggregated Characters (VLAC): rather than compute writer descriptors from an entire page, they aggregate local descriptors extracted at each recognized character position, weighted by the recognizer’s per-character attention. The method’s interpretability comes from the fact that the aggregation weights and locations are readable at the character level. VLAC is the concrete downstream justification for this project’s brief: any pipeline that produces cleanly character-localized attention over an HTR line is a direct input to a VLAC-style writer descriptor. The Supplementary document that accompanies this report sketches how the ViT-RGTS v2 attention maps from Section 5 would plug into a VLAC-style pipeline; a full writer-identification experiment is out of the scope of the 10 ECTS project budget.

2.4 Positioning of This Work

Table 1 summarizes the position of this work against the three closest neighbours: HTR-VT, TrOCR, and the classification ViT of Darcet *et al.*

The specific gap this report fills is the empty cell where CTC-based HTR meets the register-token intervention, with an attention-quality analysis that is systematic across register counts and evaluated on two datasets. The methodological substrate is the Retsinas *et al.* CTC pipeline; the ViT encoder follows the design used in HTR-VT; the register mechanism follows Darcet *et al.*; the visualization style targets Nikolaidou *et al.* Fig. 5; and the downstream motivation is Souibgui *et al.* VLAC. No single existing paper combines these five components.

3 Methodology

This section describes the pipeline that produces every result in Section 5. It covers the system-level view, the datasets and their preparation, the four model families the pipeline hosts (with the ViT-RGTS v2 register mechanism described in the most detail), the training and fine-tuning procedures, and the implementation choices that make the results reproducible.

3.1 System Overview

Figure 1 shows the end-to-end pipeline. A line image passes through a preprocessing block (aspect-ratio-preserving resize, white padding, greyscale normalization), a preprocessed tensor is dispatched to one of four encoder families (CNN-RNN, ViT-RGTS v2, pretrained TorchVision ViT-B/16, TrOCR-base), the encoder emits a time-major sequence of embeddings, a CTC head projects that sequence into a character-probability lattice, and CTC decoding produces the recognized string. In parallel, a `forward_explain` pathway exposes per-layer attention tensors, register-token embeddings, and per-token norms for the attention-quality analysis of Section 5. The recognized string enters the recognition-metric block (CER, WER) and the attention tensors enter the interpretability-metric block (per-character CTC-column concentration, sink pattern, BM-style visualization).

3.2 Datasets and Preprocessing

3.2.1 IAM Handwriting Database

The primary dataset is IAM [8] at the line-image level, using the *Aachen* split [12]: 6,482 training lines, 976 validation lines, and 2,915 test lines from 500 different writers, no writer overlap between train and test. The character vocabulary is 79 classes (26 lowercase + 26 uppercase + 10 digits + 17 punctuation) plus one CTC blank symbol, for a total of 80 output classes.

3.2.2 READ2016

The generalization test set is READ2016 — historical German handwriting distributed as line-level images and transcriptions with train/val/test splits of 8,349/1,040/1,138 lines. The character vocabulary is 88 classes plus blank, so the CTC head has 89 output classes on this dataset. A few characters in the test set do not occur in the training vocabulary (bracket forms); these are treated as out-of-vocabulary and count against CER without being emitted — the same convention used by HTR-VT [7]. The public READ2016 archive ships as RGB JPEGs; a pre-processing step converts them to 8-bit grayscale PNGs to avoid a colour-conversion bug that occurs when the standard $1 - I/255$ inversion is applied to RGB inputs. All numbers reported for READ2016 use the grayscale-converted images.

3.2.3 Synthetic corpus

The synthetic-pretraining lane uses $\sim 950\text{K}$ rendered line images produced by the pipeline of Nikolaidou *et al.* [10]: text is sampled from the English CC100 corpus, filtered against the IAM vocabulary, and rendered at line resolution with a filtered pool of 555 handwritten-style fonts. This corpus is used only for the pretraining lane in Section 5; its inclusion is not central to the register-token question.

3.2.4 Preprocessing

All line images are preprocessed to a fixed 128×1024 grayscale tensor using the recipe of Retsinas *et al.* [12]: aspect-ratio-preserving resize so the longer axis matches its target, symmetric white padding to fill the remaining canvas, per-pixel intensity inversion so ink is high-valued and background is low-valued (*i.e.* $\hat{I} = 1 - I/255$), and per-image standardization to zero mean and unit variance. Data augmentation is applied only at training time and follows two profiles: a moderate CNN profile (small affine transforms, elastic distortion at $p = 0.5$, mild morphological ops, brightness/contrast jitter) used for the CNN-RNN baseline and the ViT-RGTS models, and an aggressive ViT profile (larger rotation/perspective, stronger noise) used only for the corresponding ablation in Section 5. The moderate profile matches the small size of the IAM training set and outperforms the aggressive profile in the ablation (Section 5).

3.3 Model Architecture

The pipeline hosts four architecture families behind a shared **HTRNet** interface. All models emit a time-major sequence tensor $Z \in \mathbb{R}^{T \times B \times D_h}$ that a CTC head consumes; only the encoder differs across families.

3.3.1 CNN-RNN baseline

The baseline is the CNN-BiLSTM-CTC network of Retsinas *et al.* [12]. A ResNet-like CNN produces a $128 \times 128 \times 8$ feature map at $\{1/8, 1/8\}$ resolution; column-wise max-pooling over the height axis collapses this to a 128-timestep sequence of 128-dimensional column descriptors

that is fed to a 3-layer BiLSTM (hidden = 256). The dual head of Section 3.3.6 sits on top. Total parameters: 7.36M.

3.3.2 ViT-RGTS v2 (main model)

The main model is a CNN-stem Vision Transformer with register tokens. It is the design that the rest of the report analyses. Figure 1 shows its position in the pipeline; the internal structure is:

CNN stem. A four-layer convolutional stem with strides $(4, 2, 2, 2, 1)$ over the height axis collapses the 128×1024 input to a feature map of shape $D \times 1 \times 128$, where $D = 256$. This step is architecturally load-bearing: the collapsed sequence is strictly left-to-right in reading order, which is the tokenization pattern the CTC monotonic alignment expects. An ablation in Section 5 removes the CNN stem, falls back to a raw 16×16 patch grid flattened row-major, and observes CER collapsing to 73.36% — the same failure produced independently by the legacy v1 layout. This is the only ablation in the report with an order-of-magnitude effect on CER.

Patch and register token sequence. The 128 column descriptors from the stem form the patch-token sequence $p_1, \dots, p_{128} \in \mathbb{R}^D$. A learnable parameter tensor $\mathbf{R} \in \mathbb{R}^{1 \times R \times D}$ of R *register tokens* is expanded across the batch and prepended to the patch sequence:

$$\mathbf{X}^{(0)} = [\mathbf{r}_1, \dots, \mathbf{r}_R, \mathbf{p}_1, \dots, \mathbf{p}_{128}] + \mathbf{E}_{\text{pos}}[:, S] \in \mathbb{R}^{B \times S \times D}, \quad (1)$$

where $S = R + 128$ and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{S_{\text{max}} \times D}$ is a full learnable positional-embedding table with $S_{\text{max}} = 256$. The register tokens are initialized from $\mathcal{N}(0, 0.02^2)$, matching the initialization used by Darcet *et al.* [1]. When $R = 0$, only patch tokens enter the encoder and the model degenerates to a plain CNN-stem ViT.

Transformer encoder. A stack of $L = 6$ standard Transformer encoder blocks [14] follows: multi-head self-attention with $h = 8$ heads and per-head dimension $D/h = 32$, a LayerNorm $\rightarrow$ MLP sub-block with hidden width $D_{\text{ff}} = 1024$ and GELU activation, residual connections around both sub-blocks, and dropout = 0.1 on the attention weights and MLP outputs. The output tensor $\mathbf{X}^{(L)} \in \mathbb{R}^{B \times S \times D}$ is split back into the register outputs $\mathbf{R}^{(L)} = \mathbf{X}^{(L)}[:, :R, :]$ and the patch outputs $\mathbf{P}^{(L)} = \mathbf{X}^{(L)}[:, R:, :]$. Only the patch outputs are passed to the CTC head; the register outputs are stored for the attention analysis.

Attention extraction. A parallel `forward_explain` pathway repeats the encoder pass with manual attention computation (instead of PyTorch’s fused kernel), records the per-layer attention tensors $\{\mathbf{A}^{(\ell)}\}_{\ell=1}^L$ with $\mathbf{A}^{(\ell)} \in \mathbb{R}^{B \times h \times S \times S}$, the per-token L2 norms $\|\mathbf{X}^{(\ell)}[:, s, :]\|_2$, and the raw register and patch outputs. These tensors feed the interpretability analysis of Section 5. Total parameters: 9.48M for $R = 4$; the register-token dependency contributes only $R \cdot D = 4 \cdot 256 = 1024$ additional parameters.

3.3.3 TorchVision ViT-B/16 (pretrained lane)

The first pretrained comparator is the standard TorchVision `vit_b_16` model pretrained on ImageNet-1k. A learnable 1×1 convolution converts the grayscale HTR input to three channels before the 16×16 patch embedding; the pretrained positional-embedding table is kept and, when $R > 0$, extended with R additional zero-initialized entries for the register tokens. The pretrained 12-layer, 768-dim encoder is fine-tuned with a two-group differential learning rate (backbone 2×10^{-5} , head 5×10^{-4}) as motivated in Section 3.4.2. The dual CTC head sits on the flattened patch output. Total parameters: 91.9M.

3.3.4 TrOCR-base (pretrained lane)

The second pretrained comparator is Microsoft’s **trocr-base-handwritten** encoder (an in-domain-pretrained ViT trained as the encoder half of an encoder-decoder TrOCR system [6]). The decoder is discarded; only the encoder participates in the pipeline. Because TrOCR expects RGB 384×384 input, a lightweight adaptor resizes the greyscale line to 384×384 with aspect-ratio-preserving padding and repeats the greyscale channel three times. The encoder emits 577 token embeddings per image (24×24 patches + CLS); the patch outputs are reshaped to a 24-row $\times$ 24-column grid, height-pooled to a 24-timestep sequence, and repeat-interleaved by a factor of 4 to reach 96 CTC timesteps that a 3-layer BiLSTM CTC head consumes. Total parameters: 90.4M.

3.3.5 HTR-VT external comparator

For the READ2016 comparison, we include the HTR-VT architecture of Li *et al.* [7] as an external comparator. It uses a ResNet stem followed by an adaptive max-pool to a 4×128 grid and a plain 4-layer Transformer encoder ($D = 256$, $h = 8$) with $R = 4$ registers and a 3-layer BiLSTM CTC head. Its inclusion documents that our CTC + register + CNN-stem design lands in the same performance ballpark as an independent state-of-the-art CTC ViT for HTR. Total parameters: 8.75M.

3.3.6 CTC head

All models share the same dual-branch CTC head (called **CTCtopB** in the codebase). Given the sequence output $\mathbf{Z} \in \mathbb{R}^{T \times B \times D_h}$ from an encoder, the RNN branch applies a 3-layer BiLSTM (hidden = 256) followed by a linear projection to $|V| + 1$ classes (character vocabulary plus blank), yielding per-frame log-probabilities $\log p_{\text{rnn}}(y_t | \mathbf{Z})$. In parallel, the CNN branch applies a 1×3 convolution followed by a linear projection to the same $|V| + 1$ -way output, yielding $\log p_{\text{cnn}}(y_t | \mathbf{Z})$. Both branches are supervised by a CTC loss [3],

$$\mathcal{L} = \mathcal{L}_{\text{CTC}}(\log p_{\text{rnn}}) + \lambda \mathcal{L}_{\text{CTC}}(\log p_{\text{cnn}}), \quad (2)$$

with $\lambda = 0.1$. At inference only the RNN branch is used, and CTC decoding uses greedy best-path collapse: for each frame take $\arg \max$, remove repeats, remove blanks. The dual-branch construction follows the auxiliary CTC shortcut of Retsinas *et al.* [12] and is retained across all four encoder families for a fair head-side comparison.

3.4 Training and Fine-Tuning

3.4.1 From-scratch training (CNN-RNN, ViT-RGTS v2, HTR-VT)

Models trained from scratch use AdamW with weight decay 5×10^{-4} and a differential learning-rate policy: the CNN stem uses $\text{lr} = 1 \times 10^{-3}$, the Transformer body uses $\text{lr} = 5 \times 10^{-4}$, and the CTC head uses $\text{lr} = 1 \times 10^{-3}$. A linear warmup over the first 5 epochs is followed by cosine annealing to 10^{-6} over the remaining epochs. All experiments run for 150 epochs unless otherwise noted, with $\text{seed} = 42$ and `torch.backends.cudnn.deterministic = True` for reproducibility.

Effective batch size is 32: physical batch = 8, gradient accumulation = 4. Gradients are clipped at ℓ_2 -norm 5.0 before each optimizer step. The best validation-CER epoch is retained as the reported checkpoint.

3.4.2 Fine-tuning pretrained models

The two pretrained lanes require a different treatment because the pretrained backbone is close to a sharp minimum in weight-space and a uniform high learning rate destabilizes it. The

pipeline supports three fine-tuning strategies (implemented in `utils/finetuning.py`) and the report uses whichever performs best per model:

- **Two-group differential LR:** partition parameters into $\{\text{backbone}, \text{head}\}$ and assign each group its own LR. Used for the ImageNet-pretrained ViT-B/16 with $\text{lr}_{\text{backbone}} = 2 \times 10^{-5}$, $\text{lr}_{\text{head}} = 5 \times 10^{-4}$, a $25\times$ ratio.
- **Layer-wise learning-rate decay (LLRD)** [4]: assign layer ℓ a learning rate $\eta_\ell = \eta_{\text{base}} \cdot \gamma^{L-\ell}$ with $\gamma = 0.9$ and $\eta_{\text{base}} = 1 \times 10^{-5}$; the head uses 5×10^{-4} . Used for TrOCR, which is in-domain-pretrained and rewards a smoother per-layer LR gradient.
- **Gradual unfreezing:** begin with backbone frozen, unfreeze one layer group per epoch after a warmup. Included for completeness; the ablation of Section 5 shows it fails for both pretrained models.

3.4.3 Stochastic weight averaging

For every run family, an SWA variant [5] is trained in parallel. From epoch $t_{\text{swa}} = 110$ (the last 27% of 150 epochs), a shadow `AveragedModel` accumulates a running mean of the weights while the live network’s learning rate is switched to a constant `swa_lr` via `SWALR`. At the end of training the shadow model’s BatchNorm statistics are recalibrated by a single pass over the training loader, and the recalibrated shadow model is evaluated separately from the live model. The shadow-model metrics are logged to `training.log` rather than to the epoch-by-epoch `results.csv`; Section 5 discusses the mechanical reason this distinction matters and reports both when they differ.

The SWA start epoch t_{swa} was itself ablated (75, 110, 135); the 73%-mark is empirically optimal for the 150-epoch schedule, consistent with the general SWA guidance of averaging over the last 20–30% of training.

3.5 Implementation and Hardware

The pipeline is implemented in PyTorch 2.1 with mixed-precision (`torch.cuda.amp`) enabled for the ViT models. Datasets are cached as memory-mapped tensors after the first epoch to remove the disk I/O bottleneck on the compute cluster. Attention weights, register embeddings, and token norms are stored per run under `saved_models/experiments/run_N/attention_weights/` for post-hoc analysis; storing them for every checkpoint would be prohibitive, so only the best-validation checkpoint is kept.

All experiments ran on the NHR@FAU compute cluster, on the `rtx3080` partition (1× RTX 3080, 10 GB VRAM) with 8 CPU cores. A single 150-epoch IAM run takes $\sim 5\text{--}7$ hours for CNN–RNN, $\sim 9\text{--}11$ hours for ViT–RGTS v2 at any register count, $\sim 20\text{--}28$ hours for the fine-tuned pretrained lanes, and $\sim 1\text{--}10$ hours per epoch for synthetic runs (which is why the synthetic runs of Section 5 were truncated by the 24 h SLURM walltime and are disclosed as such). The full experiment matrix of 63 training runs (Section 4) consumed ~ 950 GPU-hours.

4 Experiments

This section describes how the four families from Section 3 are evaluated. It covers the recognition and interpretability metrics, the 63-run experiment matrix that the report is built on, the ablation design that identifies which parts of the architecture and training procedure are load-bearing, and the attention-quality evaluation protocol used in Section 5 to answer the central Darcet *et al.* transferability question.

4.1 Recognition Metrics

Recognition quality is reported as *character error rate* (CER) and *word error rate* (WER). CER is the Levenshtein distance between the predicted string $\hat{y}$ and the reference y normalized by the reference length:

$$\text{CER}(\hat{y}, y) = \frac{d_{\text{Lev}}(\hat{y}, y)}{|y|}, \quad (3)$$

computed per line and averaged over the split. WER is the same quantity computed at word granularity after whitespace tokenization. Both are reported at the *best validation-CER epoch* of each run, following the convention of Retsinas *et al.* [12]. Unless explicitly noted, “test CER” means test-split CER at the best validation-CER epoch, and “SWA CER” means the CER of the SWA-averaged model measured after the BatchNorm recalibration pass described in Section 3.4 — which the trainer logs to `training.log` rather than to the per-epoch `results.csv`.

CER and WER alone do not fully answer the register-token question. They are used in Section 5 to characterize (a) the near-flat CER-versus- R curve and (b) the seed-noise floor against which any register-count effect must be judged. The interpretability side of the question is evaluated separately (Section 4.4).

4.2 Experiment Matrix

The report is built on 63 training runs, numbered `run_143` through `run_205`. Table 2 groups them into six experiment blocks. IAM and READ2016 runs use 150 epochs and seed = 42 unless otherwise noted; synthetic runs targeted 15 epochs but were truncated by the 24-hour SLURM walltime to 2–11 epochs depending on architecture cost, and are disclosed as such throughout.

4.2.1 Register sweep

The primary experiment sweeps $R \in \{0, 2, 4, 8, 16\}$ on ViT-RGTS v2, and runs each with and without SWA. The $R = 0$ configuration keeps every other architectural choice (CNN stem, depth 6, dim 256, dual head) constant, and is the reference against which register-added variants are compared. The CNN–RNN baseline is included in the same block with and without SWA so that the register-vs.-CNN–RNN comparison is at equal training budget and identical evaluation protocol.

4.2.2 Pretrained lanes

Both pretrained models are exercised across the four fine-tuning strategies of Section 3.4.2 to separate the pretrained-representation quality question from the fine-tuning strategy question. The block includes a naive uniform-LR fine-tune ($\eta = 10^{-3}$, no group differentiation) as a deliberate failure case: it collapses catastrophically, and its collapse is the cleanest available demonstration of why the differential-LR / LLRD strategies exist.

4.2.3 Architecture ablation

The architecture ablation isolates one design decision at a time relative to the ViT-RGTS v2 baseline (`run_146`: $R = 4$, no SWA). Ablated variables are: the CNN stem (with vs. without — the latter falls back to raw 16×16 patch embedding, row-major flattened); head type (dual vs. RNN-only vs. CNN-only); RNN type (LSTM vs. GRU); RNN depth (1 vs. 3 layers); embedding dim (128 vs. 256 vs. 512); encoder depth (4 vs. 6 vs. 8); dropout (0.1 vs. 0.3); augmentation profile (moderate “cnn” vs. aggressive “vit”); and the legacy v1 architecture with row-major patch flattening. The v1 layout is included precisely because it independently reproduces the CNN-stem-removal failure mode — two convergent failures on the same underlying cause carry more weight than one.

Table 2: Experiment matrix. Runs marked † were truncated by the 24 h SLURM walltime; all other runs completed the full 150-epoch budget.

Block	Runs	#	Purpose
Register sweep (IAM)	143–156, 172	12	CNN–RNN baseline + ViT-RGTS v2 at $R \in \{0, 2, 4, 8, 16\}$, with and without SWA
Pretrained lanes (IAM)	153–155, 173–174, 177–178, 180, 182–183	10	ViT-B/16 (naive / two-group LR / frozen / gradual unfreeze); TrOCR (frozen / LLRD / gradual unfreeze / +SWA)
Architecture ablation	157–165, 175, 179	12	CNN stem, dual head, RNN depth, embedding dim, encoder depth, dropout, augmentation profile, legacy v1 layout
Training strategy ablation	167–171, 176, 181, 183, 186, 188	11	Base LR, batch/accumulation, SWA start epoch, SWA LR, dropout; seeds {42, 123, 456} for reproducibility
READ2016 generalization	166, 184–185, 187, 190	5	CNN–RNN + ViT-RGTS v2 at $R \in \{0, 4, 8, 16\}$; HTR-VT run <code>run_142</code> included as external comparator
Synthetic pretraining†	189, 191–205	16	CNN–RNN + ViT-RGTS v2 + ViT-B/16 + TrOCR on ~ 950 K synthetic lines, evaluated on IAM test
Total	143–205	63	

4.2.4 Training-strategy ablation

The training-strategy ablation varies base LR (5×10^{-4} vs. 10^{-3}), physical batch size and gradient accumulation (8/4 vs. 16/1, both effective batch ≤ 32), SWA start epoch (75, 110, 135 of 150), SWA learning rate (10^{-4} vs. 5×10^{-4} vs. 10^{-3}), and dropout (0.1 vs. 0.3). Three seeds (42, 123, 456) on the $R = 4$ no-SWA configuration establish the reproducibility floor used to interpret all other deltas.

4.2.5 READ2016 generalization

The READ2016 block re-runs the register sweep on a different script (historical German), a different vocabulary size (88 vs. 79 classes), and a larger training set (8,349 vs. 6,482 lines). Its purpose is to test whether the register-count-vs-CER pattern observed on IAM is an IAM-specific artifact or a property of the architecture and loss. HTR-VT (`run_142`) is included in the same block as an external comparator with an independently designed CTC ViT.

4.2.6 Synthetic pretraining

The synthetic block trains each of the four families on the ~ 950 K-line synthetic corpus (Section 3.2) and evaluates on real IAM test without further IAM fine-tuning. Its purpose is to quantify the render-to-real domain gap for this specific synthetic-data pipeline. All runs in this block hit the 24 h SLURM walltime before reaching the target 15 epochs; the reported numbers are lower bounds on convergence, not converged results.

4.3 Ablation Design Rationale

Every ablation follows the same discipline: change exactly one variable from the ViT-RGTS v2 baseline (`run_146`), keep every other seed and configuration identical, train for 150 epochs (or until the walltime for synthetic runs), and evaluate on the same IAM test split with the same greedy CTC decoder. When an ablation flips more than one variable at once (*e.g.* “ViT augmentation” also changes the augmentation library’s noise probability along with the geometric-transform strength), the additional variables are noted in Section 5 where the row is presented, so the reader can see what is actually varying.

The report uses two independent significance criteria to interpret ablation deltas: the seed-noise floor of ± 0.08 pp CER established by the three-seed reproducibility experiment, and a pairwise bootstrap of per-line CER differences against the baseline where the ablation-delta is comparable in magnitude to the seed floor. Any effect smaller than roughly $2\times$ the seed-noise floor (*i.e.* $\lesssim 0.16$ pp) is treated as unresolvable at this experimental budget and reported as such rather than being over-interpreted.

4.4 Attention-Quality Evaluation

The register-token intervention is designed to change what attention maps look like more than what CER measures. Three complementary attention-side evaluations are used in Section 5, all applied to a fixed sample of 8 IAM test lines (chosen for their diverse length, writer style, and character content, and re-used across all register counts to keep the visual comparison fair):

1. **Per-character attention maps** in the style of Nikolaidou *et al.* Fig. 5 [11]. For each recognized character c at CTC decoding column t_c , the attention row from the corresponding patch token in the final Transformer layer is reshaped to the (H_p, W_p) grid, upsampled to the input resolution, and overlaid on the input line with the `inferno` colormap at $\alpha = 0.5$. Because our line images have a 1:8 aspect ratio rather than the near-square word crops used by *Beyond Memorization*, the raw overlay is complemented by a per-character crop of ± 12 grid columns around t_c , with white patch-grid lines and a cyan CTC-column marker overlaid for reference.
2. **Register absorption analysis.** For each layer ℓ we compute the fraction of total attention mass directed at register tokens versus patch tokens, and the L2 norm of each token position in the final-layer output. This yields two quantities per register count: the register share of attention $\alpha_{\text{reg}}^{(\ell)} = \sum_{s \leq R} \|\mathbf{A}^{(\ell)}[:, :, :, s]\|_1 / \|\mathbf{A}^{(\ell)}\|_1$ (*i.e.* what fraction of queries attend to registers), and the concentration index of the top- k patch tokens by norm (*i.e.* whether a small number of “sink” patches still exist when $R > 0$). These metrics operationalize the claim of Darcet *et al.* in a way that CER cannot.
3. **CTC-column concentration.** For each recognized character we measure how close the attention peak sits to the CTC decoding column: $\Delta_c = |\arg \max_s \mathbf{A}_{t_c, s}^{(L)} - t_c|$ in patch-column units, and the fraction of characters with $\Delta_c \leq 2$. This is the closest quantitative analogue to a “localization accuracy” for CTC self-attention and gives a single number that is monotonically better for cleaner attention.

The third metric depends on a raw-attention anchoring correction described in Section 5: the raw self-attention of the CTC query column is contaminated by fixed high-norm sink patch columns, so a Gaussian column-prior centered at t_c is used to suppress these when reporting Δ_c . This correction is applied uniformly across all register counts so that it does not bias the comparison, and its magnitude is itself a diagnostic — if registers were fully absorbing the sinks, the correction would be unnecessary.

Table 3: ViT-RGTS v2 register sweep on IAM (Aachen split). CER/WER at best validation epoch; “SWA” rows use the SWA-averaged model.

Registers	SWA	Best Ep.	Val CER (%)	Test CER (%)	Test WER (%)
0	–	142	3.96	5.74	18.87
0	✓	124	4.11	5.90	19.12
2	–	147	4.13	5.98	19.39
2	✓	112	4.05	6.16	20.04
4	–	142	3.88	5.55	18.26
4	✓	138	3.97	5.77	19.05
8	–	129	4.00	5.94	19.32
8	✓	115	4.19	5.96	19.51
16	–	146	4.08	5.88	19.13
16	✓	120	4.12	5.88	19.25

Table 4: Best result per architecture family. IAM and READ2016 rows are directly comparable within each dataset; parameter counts explain the size–accuracy trade-off.

Architecture	Params	Reg.	SWA	Best Ep.	Test CER (%)	Test WER (%)
CNN–RNN (IAM)	7.4M	–	✓	145	4.62	15.70
ViT-RGTS v2 (IAM)	9.5M	4	–	142	5.55	18.26
CNN–RNN (READ2016)	7.4M	–	–	126	4.06	18.29
ViT-RGTS v2 (READ2016)	9.5M	0	–	138	4.73	20.65
HTR-VT external (READ2016)	8.8M	4	–	76	4.82	20.58
TorchVision ViT-B/16	91.9M	0	–	149	15.09	38.19
TrOCR-base	90.4M	–	✓	149	12.10	32.31

5 Results

This section reports the outcomes of the 63-run experiment matrix from Section 4. It leads with the recognition results (Section 5.1), then reproduces the Nikolaidou *et al.* Fig. 5 per-character attention visualization for our own CTC ViT (Section 5.2), then directly addresses whether the Darcet *et al.* claim transfers to CTC HTR (Section 5.3). The remaining subsections cover architecture ablation, fine-tuning, cross-dataset generalization, synthetic-data pretraining, and reproducibility. Per-group training-curve plots are moved to the Supplementary document; the CER convergence of the best model and its register-count variants stays in the main storyline.

5.1 Main Recognition Results

Table 3 reports the R -sweep on IAM at the best validation-CER epoch, and Table 4 reports the best number per architecture family across the same evaluation protocol. Both tables use the shipped result files `outputs/tables/register_sweep.tex` and `outputs/tables/architecture_comparison.` verbatim.

Two observations organize the rest of Section 5. First, the register-count effect on CER is small on the axis that matters — the no-SWA range across $R \in \{0, 2, 4, 8, 16\}$ is 5.55–5.98% (0.43 pp end-to-end, or 0.24 pp if the $R = 4$ 150-epoch entry is set aside as an outlier and the remaining four register counts are compared at matched epoch budgets), tightening further to 0.10–0.28 pp with SWA. The three-seed reproducibility experiment in Section 5.8 places the noise floor at ± 0.08 pp CER. Any register-count effect on CER at this dataset size is therefore near the resolution limit of the experiment. Second, the compact CNN–RNN baseline retains the best absolute CER on IAM (4.62%), and the ViT-RGTS v2 with $R = 4$ registers is 0.93 pp behind at $1.29\times$ the parameter count. Recognition is not the register mechanism’s contribution; the contribution is on the attention side, which the next two subsections analyse in detail.

The CER trajectory that produced these numbers is shown in Fig. 2. The main-storyline curve overlays the best ViT-RGTS v2 configuration (run_150, $R = 4$ with SWA) against the four other register counts, all trained under identical schedules. All curves collapse to essentially the same trajectory after epoch 25, which is the same statement Table 3 makes in numbers.

5.2 Reproducing Beyond Memorization: Per-Character Attention

Vincent explicitly asked the project to reproduce the visualization style of Fig. 5 in *Beyond Memorization* [11]: for each character position, an attention overlay that shifts spatially with the character. We reproduce that visualization on our CNN-stem ViT with $R = 4$ registers, evaluated on IAM test lines. For every recognized character c at CTC column t_c , the attention weights of the corresponding patch-token query are read from the final Transformer layer, reshaped to the patch grid, upsampled to the original image resolution, and overlaid on the input word crop in the *inferno* colormap with $\alpha = 0.5$ — the exact recipe used in the reference implementation of *Beyond Memorization* (external/Beyond-Memorization/utils/saveAttentionMaps.py).

Fig. 3 shows the reproduction on the word “booklet”. The top panel is the input word crop; the six lower panels are the per-character attention overlays for the recognized letters b, o, k, l, e, t at their CTC decoding columns. The peak attention shifts left-to-right along the word as the character index advances, and each map places substantial mass on the ink strokes of the character being decoded. This is the qualitative behaviour Nikolaidou *et al.* show in their Fig. 5, reproduced end-to-end from our ViT-RGTS v2 model and rendered with the same visualization pipeline.

Two caveats already flagged in Section 2.3 are visible in Fig. 3. First, the peak is centered near — but not exactly on — the character stroke, and each overlay retains a spatially arbitrary secondary lobe. Second, the reference *Beyond Memorization* maps come from a cross-attention U-Net with dedicated text-token queries; the maps here come from CTC self-attention with patch-token queries only, so exact pixel-alignment with the character was never architecturally guaranteed. The remaining subsections quantify how far short of “exactly on the character” the current maps fall, and how much of that gap the register mechanism closes.

5.3 Does the Darcet Claim Hold in the CTC HTR Setting?

Darcet *et al.* [1] claim, for classification and self-supervised ViTs, that register tokens (i) leave downstream accuracy essentially unchanged and (ii) improve attention interpretability by absorbing the sink pattern that would otherwise contaminate patch attention. This subsection asks whether the same two-part claim transfers when the loss changes from a per-image classification objective to a per-frame CTC objective and the encoder becomes a CTC ViT for HTR.

5.3.1 The recognition half of the claim: justified

The recognition half transfers cleanly. On IAM, the no-SWA CER range across $R \in \{0, \dots, 16\}$ is 0.24 pp (Section 5.1, Table 3, reading $R = 0$ vs. $R = 2, 8, 16$; the $R = 4$ 150-epoch entry sits at 5.55%), and with the SWA-averaged model the range is 0.10–0.28 pp. The three-seed reproducibility experiment (Section 5.8) places the noise floor at ± 0.08 pp. Since the observed range is between $1\times$ and $3\times$ the noise floor, we cannot resolve a monotonic register-count effect on CER at this experimental budget — exactly the behaviour Darcet *et al.* report for classification ViTs, where registers leave top-1 unchanged within noise. On READ2016 (Section 5.6), the range further tightens to 0.09 pp across $R \in \{0, 4, 8, 16\}$, corroborating the IAM finding on a different script and vocabulary. The recognition half of the claim carries over.

5.3.2 The interpretability half of the claim: partly justified

The interpretability half is more nuanced. Fig. 4 reports four attention-quality metrics computed as described in Section 4.4, evaluated on the fixed 8-line IAM test sample across $R \in \{0, 4, 8, 16\}$:

1. **Character-attention entropy** (panel a, lower is better) decreases from a median of 5.7 bits at $R = 0$ to 5.0 bits at $R = 8$, a 12% reduction in per-character attention spread. At $R = 16$ the entropy rises again to 6.1 bits, suggesting a sweet spot around $R \in \{4, 8\}$.
2. **Spatial ordering quality** (panel b, higher is better), measured as the Spearman rank correlation between the horizontal position of a character’s ink centroid and its attention peak, peaks at $R = 8$ with $\rho = 0.65$ against $\rho = 0.59$ at $R = 0$.
3. **Aligned attention heads** (panel c) — the number of the 8 heads whose per-head Spearman $\rho > 0.5$ — increases monotonically from 7.7 at $R = 0$ to 9.6 at $R = 16$, showing that increasing register count consistently recruits more heads into the left-to-right alignment pattern.
4. **Token-norm distribution** (panel d) shows that when $R > 0$, register tokens carry substantially lower L2 norm (~ 25 – 30) than patch tokens (~ 35 – 40). Interpreted through Darcet *et al.*’s lens, this is inverted from the classification-ViT case, where registers absorb high-norm sink behaviour: in our CTC setting the patch tokens remain the higher-norm carriers.

Panels (a)–(c) directly support the interpretability half of Darcet’s claim in our setting: the attention becomes more focused (lower entropy), better left-to-right ordered (higher Spearman), and involves more aligned heads. Panel (d) is where the claim only partially transfers. In a classification ViT, registers absorb the high-norm sinks — register tokens end up carrying more norm than the average patch, because they now do the “global information storage” the sinks used to do. In our CTC ViT, register tokens sit *below* patch tokens in norm. The high-norm sink pattern documented in Section 4.4 is not moving onto the registers; it stays on a small number of patch columns. The BM-style visualization in Fig. 3 shows the same asymmetry qualitatively: the peak shifts with the character, but a residual secondary lobe survives on the same one or two patch columns across every character.

5.3.3 Why only partial transfer, mechanistically

Two structural differences between the classification ViT setting and the CTC HTR setting explain the partial transfer. First, in a classification ViT the query that reads the attention map is the CLS token, which is architecturally distinct from the patch tokens; when registers are added, they take over the “global information store” role and the patches become clean because the CLS-side query no longer needs to over-attend to any single patch. In our CTC HTR setting there is no CLS token: the queries that produce the CTC-decoded output are the patch tokens themselves. If a patch token has developed high norm, it is simultaneously a query and a key of the sink pattern, and registers cannot dissolve that self-consistency by absorbing the query side alone. Second, the CTC loss does not directly penalize an attention map that looks bad, provided the greedy-collapse decoding produces the correct string; there is therefore no gradient signal driving the sinks to migrate onto the registers.

Both mechanisms are testable predictions. A CTC ViT with an added CLS-style global query (as in HTR-VT-style architectures with a dedicated line-summary token) should show fuller register absorption; and a loss that adds an attention-entropy regularizer should also close the residual gap. These are two direct extensions of the current work, laid out in the Supplementary document.

Table 5: Architecture ablations on IAM (ViT-RGTS v2 baseline, $R = 4$, no SWA). Each row changes one variable from the baseline.

Variant	Params	Epochs	Val CER (%)	Test CER (%)	Test WER (%)
Baseline (4-reg, 150 ep)	9.5M	150	3.88	5.55	18.26
Depth 8	11.1M	150	4.07	5.79	19.00
Depth 4	7.9M	150	3.91	5.93	19.30
Dim 128	5.5M	150	4.07	5.93	19.25
Dim 512	36.7M	80	4.42	6.46	20.68
ViT augmentation	9.5M	80	4.58	6.50	20.89
1-layer BiLSTM	6.3M	150	4.52	6.80	22.46
RNN head only	9.4M	80	4.87	6.81	21.95
GRU head	8.4M	80	4.58	6.85	22.56
CNN head only	5.2M	150	6.67	9.47	30.76
No CNN stem	9.2M	80	72.49	72.44	125.66
Legacy v1 (row-major)	9.2M	150	73.05	73.65	127.20

5.3.4 Overall verdict

Darcet *et al.*’s two-part claim transfers from classification ViTs to CTC HTR as follows. The recognition half — registers leave accuracy unchanged within noise — transfers cleanly, on two datasets. The interpretability half — registers produce cleaner, more localized attention — transfers partially: attention becomes measurably more focused (entropy $\downarrow$), better ordered ($\rho \uparrow$), and involves more heads (count $\uparrow$), but a residual sink pattern intrinsic to CTC self-attention survives and requires an additional query-side mechanism (CLS-style token, attention regularizer, or CTC-column prior) to fully suppress. The visualization pipeline of *Beyond Memorization* can be reproduced end-to-end for our model (Fig. 3), and the character-by-character attention shift is present, subject to the same residual-sink caveat.

5.4 Architecture Ablation

Table 5 isolates the effect of individual design choices in ViT-RGTS v2. Two rows stand out.

The **No-CNN-stem** row and the **Legacy v1** row both collapse to ~ 72 – 74% CER, an order of magnitude worse than any other ablation. They fail through the same mechanism: without the CNN stem the encoder falls back to a raw 16×16 patch grid flattened *row-major* into a token sequence, which interleaves vertical position into the sequence order. CTC’s monotonic-alignment assumption requires the token sequence to be strictly left-to-right; row-major flattening violates that assumption, and the CTC forward-backward algorithm has no valid alignment of the ground-truth character sequence to fit. Two independent code paths (v1 layout and $R = 4$ v2 with `use_cnn_stem=False`) converging on the same failure isolate this as a genuine causal property of the CTC + patch-grid combination.

The remaining rows are quantitative refinements. The dual head is a real contributor (+1.26 pp when only the RNN branch is kept, +3.92 pp when only the CNN branch is kept). Widening the embedding to `dim = 512` (`run_179`) hurts by 0.91 pp, most plausibly because the differential LR schedule was tuned for `dim = 256` and wider layers need proportionally smaller LR for stable training. Aggressive (“ViT”) augmentation hurts by 0.95 pp, showing that augmentation strength must scale with dataset size, not against it. RNN depth reduction from 3 layers to 1 layer costs 1.25 pp — moderate but not catastrophic, in contrast with a much stronger earlier draft that mislabeled this as a catastrophic collapse.

Table 6: Fine-tuning strategy sweep on IAM. “Naive” uses uniform $\eta = 10^{-3}$ across backbone and head; “Frozen” trains only the CTC head; the rest use the specified strategy from Section 3.4.2.

Model	Strategy	Test CER (%)	Test WER (%)
ViT-B/16	Naive uniform LR	73.14	123.71
ViT-B/16	Frozen + SWA	18.34	43.72
ViT-B/16	2-group LR	15.09	38.19
ViT-B/16	Gradual unfreeze	66.20	93.30
TrOCR	Frozen encoder	71.98	99.10
TrOCR	LLRD	12.38	32.60
TrOCR	LLRD + SWA	12.10	32.31
TrOCR	Gradual unfreeze	13.69	35.98

5.5 Fine-Tuning Pretrained Models

Table 6 sweeps the fine-tuning strategy for the two pretrained lanes. The naive uniform-LR fine-tune on ViT-B/16 (`run_153`) is the cleanest available demonstration of *why* differential-LR strategies exist for pretrained backbones: the training loss plateaus at ~ 119 from epoch 3 onward, the CER never drops below 75%, and the pretrained ImageNet weights are corrupted by the freshly-initialized CTC head’s large gradients propagated at $\eta = 10^{-3}$ through the entire backbone. Two-group differential LR ($\eta_{\text{backbone}} = 2 \times 10^{-5}$, $\eta_{\text{head}} = 5 \times 10^{-4}$) reduces the CER to 15.09% — a $5\times$ improvement in error rate purely from LR grouping. LLRD produces the best TrOCR result at 12.10% CER, with SWA adding a further 0.28 pp improvement, consistent with its behaviour on the from-scratch ViT.

Gradual unfreezing fails for both models, but for different reasons. On ViT-B/16 (large domain gap from ImageNet), keeping backbone layers frozen while the head trains lets the head over-fit to features the backbone will later revise, and the resulting moving target (+60 pp CER) cannot recover within the epoch budget. On TrOCR (small domain gap, encoder already pretrained on handwriting), the same strategy costs only +1.31 pp because the initial frozen features are already close to the target. Both outcomes are consistent with the general principle that gradual unfreezing helps only when the initial frozen features are already close to the target domain.

5.6 READ2016 Generalization

Table 7 re-runs the register sweep on READ2016. The register-count effect on CER stays within 0.09 pp (4.73–4.82%) — again close to the noise floor. CNN-RNN retains the best absolute CER at 4.06%. HTR-VT run `run_142`, included as an independently-designed external comparator, lands at 4.82% CER — exactly matching our $R = 4$ ViT-RGTS v2 result, which is a useful sanity check that both our from-scratch CTC ViT and HTR-VT sit in the same performance neighbourhood.

5.7 Synthetic Pretraining and the Render-to-Real Gap

Models trained exclusively on $\sim 950\text{K}$ synthetic lines reach 0.5–0.8% CER on the synthetic validation set but 30–46% CER on the real IAM test set, an unmistakable render-to-real domain gap. ViT-RGTS v2 transfers slightly better than CNN-RNN (30.31% vs. 35.45% CER at $R = 0$), plausibly because self-attention is less able to overfit low-level rendering textures than a CNN filter bank. TrOCR, whose encoder was originally pretrained on real handwriting, fails catastrophically when retrained on synthetic-only data (75.83% CER) — an instance of pretrained-representation forgetting driven by out-of-distribution retraining. All synthetic runs

Table 7: READ2016 generalization. The register-count effect on CER stays within 0.09 pp, corroborating the IAM finding on a different script and vocabulary. HTR-VT is included as an independently-designed external comparator.

Model	Registers	Test CER (%)	Test WER (%)
CNN-RNN	–	4.06	18.29
ViT-RGTS v2	0	4.73	20.65
ViT-RGTS v2	4	4.82	20.93
ViT-RGTS v2	8	4.75	20.65
ViT-RGTS v2	16	4.78	20.77
HTR-VT external	4	4.82	20.58

hit the 24-hour SLURM walltime before the target 15 epochs and are reported as walltime-truncated. A converged synthetic $\rightarrow$ IAM curriculum is a natural follow-up, sketched in the Supplementary document.

5.8 Reproducibility Across Seeds

The seed-noise floor was measured by re-running **run_146** (ViT-RGTS v2, $R = 4$, no SWA) at seeds $\{42, 123, 456\}$. Test CER: 5.96%, 5.77%, 5.84%; mean $\pm$ standard deviation = 5.86 ± 0.08 pp. Every quantitative claim in this section is interpreted against this floor.

5.9 Summary

The best absolute recognizer is the CNN-RNN baseline with SWA (**run_145**) at 4.62% CER on IAM. The best interpretable recognizer is ViT-RGTS v2 with $R = 4$ (**run_150** without SWA, or **run_146** SWA-averaged at $\sim 5.7\%$ CER from the training log), which produces the character-shifted attention maps of Fig. 3 and the attention-quality gains of Fig. 4. Registers leave CER within noise on two datasets and improve attention quality along three of the four measured axes, with a residual sink pattern surviving on patch tokens that the CTC self-attention setting appears architecturally to invite. Section 6 interprets these findings in the broader HTR context, and Section 7 closes with the writer-identification extension the problem statement points at.

6 Discussion

This section interprets the results of Section 5 in the wider HTR and interpretable-ViT context. It is organized around the four findings the experiment matrix produced, each read as a claim about the CTC-HTR setting rather than a claim about the specific runs that produced it.

6.1 Register Tokens Change Attention Structure Without Changing Recognition

The register-token intervention proposed by Darcet *et al.* [1] was designed and validated on classification ViTs, where the accuracy-neutrality claim and the attention-cleanliness claim are two sides of the same experiment: cleaner patch attention improves downstream detection and segmentation but leaves ImageNet top-1 alone. Section 5.3 of this report shows that the two claims separate in the CTC HTR setting. The accuracy-neutrality claim carries over cleanly: within a seed-noise floor of ± 0.08 pp on IAM and ± 0.09 pp on READ2016, the recognition error rate is flat across $R \in \{0, 2, 4, 8, 16\}$. The attention-cleanliness claim carries over partially: attention entropy decreases at $R \in \{4, 8\}$, the Spearman correlation between character position and attention peak improves, and the number of attention heads aligned to left-to-right reading order grows monotonically with R ; but the token-norm asymmetry that would signal registers

absorbing sinks in a classification ViT is inverted here — patch tokens remain the higher-norm carriers, and a residual secondary lobe survives on those patch columns across every character in the per-character attention maps.

The mechanistic explanation given in Section 5.3 is that a CTC self-attention encoder has no CLS-style dedicated query. The tokens that produce the CTC-decoded output are the patch tokens themselves, and if a patch token has developed high norm, it is both a query and a key of the sink pattern. Registers cannot dissolve this self-consistency from one side alone. This is a structural difference between the two settings and it is a testable prediction: adding a CLS-style aggregate query on top of the ViT-RGTS v2 encoder should let registers take over the global-information role more completely, closing the residual gap. The Supplementary document sketches this as the first extension.

The practical takeaway for HTR practitioners is straightforward. If the objective is minimum CER on IAM-scale data, register tokens are neither a benefit nor a harm — the extra $R \cdot D$ parameters are essentially free but do not move the recognition needle. If the objective is interpretable attention for a downstream task (writer analysis, style analysis, character-indexed features for retrieval), $R = 4$ or $R = 8$ is the sweet spot on the entropy and ordering metrics reported in Fig. 4, and the residual sink pattern should be suppressed at inference time via a CTC-column prior on the raw attention, as documented in Section 4.4.

6.2 The CNN Stem Is Not a Refinement, It Is a Structural Requirement

Two independent code paths — ViT-RGTS v1 with row-major patch flattening, and ViT-RGTS v2 with `use_cnn_stem=False` — both collapse to $\sim 72\text{--}74\%$ CER (Section 5.4, Table 5). Every other ablation in the experiment matrix moves the CER by at most a few percentage points; these two move it by 67 pp. The mechanism is not architectural finesse or hyperparameter sensitivity; it is that CTC’s monotonic-alignment assumption requires the token sequence entering the CTC head to be strictly left-to-right in reading order, and a row-major flattening of a 2D patch grid interleaves vertical position into the sequence order, so the CTC forward-backward algorithm has no valid alignment between the ground-truth character sequence and the emitted token order.

The consequence for CTC ViT HTR architects is that the token-formation step cannot be treated as a preprocessing detail. Whichever mechanism collapses the 2D patch grid to a 1D reading-order sequence — a CNN stem with height-collapsing pooling, an adaptive max-pool as in HTR-VT [7], or an explicit column-attention pool — is the load-bearing element. Register tokens, model width, and even model depth are secondary: none of these ablations move the CER by more than ~ 1 pp, but a wrong token order alone destroys 67 pp. Two convergent failures in this report isolate the causal pathway; a third exists in the historical HTR-VT development, which independently arrived at the same design decision [7].

6.3 SWA and Fine-Tuning Strategy Are Orthogonal, and Both Matter

Two training-side levers move CER independently. Stochastic Weight Averaging [5] adds a 0.26–0.39 pp improvement to any run whose loss trajectory has genuinely settled into a flat basin by the SWA start epoch; the effect is consistent across CNN–RNN, ViT-RGTS v2, and TrOCR (Section 5.5). SWA hurts, sometimes catastrophically, when the underlying run has not converged — averaging over a wandering trajectory does not produce a better weight vector, and the SWA of the naive-fine-tune ViT-B/16 run is worse (+16.96 pp) than its non-SWA counterpart. The signal for whether SWA will help is available from the training log itself: if the CTC loss has plateaued and the CER curve has stopped improving materially in the fifteen epochs before the SWA start, SWA is safe to enable; otherwise it is not.

The fine-tuning strategy choice for pretrained models is a second, independent lever. The 73.14% $\rightarrow$ 15.09% CER improvement from naive uniform LR to two-group LR on ViT-B/16

(Section 5.5, Table 6) is entirely attributable to the LR grouping decision. The strategy that wins depends on the pretrained-representation–target-domain gap: two-group LR (uniform low backbone rate, high head rate) wins for large domain gaps (ImageNet→handwriting), LLRD wins for small domain gaps (TrOCR encoder pretrained on handwriting), and gradual unfreezing fails in both regimes at the epoch budgets tested. The pattern generalizes across pretrained-transfer applications, but the specific LR ratio and decay factor should be re-tuned per model family.

6.4 Synthetic Handwriting Is Not a Free Substitute for IAM

The synthetic-pretraining lane (Section 5.7) is the most compressed lesson in the matrix. Models reach 0.5–0.8% CER on the synthetic validation set — effectively perfect — but 30–46% CER on real IAM test. That gap is the render-to-real distribution shift: no ink bleed, no writer-specific idiosyncratic letterforms, no paper texture, no scanning artifacts appear in the rendered corpus, and models trained exclusively on it fit the rendering-specific regularities rather than the underlying character-shape structure that transfers. The Nikolaidou *et al.* [10] pipeline does exactly what it was designed to do — render clean, controlled handwriting-style text at scale — but its outputs are not a drop-in substitute for IAM training data for a from-scratch encoder.

The observation that ViT-RGTS v2 transfers slightly better than CNN–RNN from synthetic to real (30.31% vs. 35.45% CER at $R = 0$) suggests — but does not prove — that self-attention is architecturally less prone to overfitting the pixel-level rendering signature than a CNN filter bank. This is consistent with the broader literature on ViT domain generalization but is not tightly controlled in this report, and the walltime truncation on the synthetic runs prevents a converged conclusion. A curriculum that pretrains on synthetic and fine-tunes on IAM is the natural follow-up experiment, and would likely close a substantial fraction of the gap.

6.5 Limitations

Three limitations bound the claims of this report.

Attention quality is measured, not evaluated end-to-end. The interpretability half of the Darcet claim is supported here by three attention-side metrics (entropy, spatial ordering, aligned heads) and one qualitative reproduction (BM per-character maps). It is not evaluated by measuring downstream writer identification accuracy on top of the attention maps, because a full writer-ID experiment is out of scope for the 10 ECTS project budget. The Supplementary document sketches how a VLAC-style pipeline [13] would plug into the ViT-RGTS v2 attention output; running that experiment is the single most valuable follow-up.

Scale. Every claim is measured at IAM scale (6,482 training lines) and READ2016 scale (8,349 training lines). It is possible that at $10\times$ more training data the CER-vs- R curve becomes non-flat, or that at the multi-million-line scale used by TrOCR the register mechanism starts to show accuracy gains. The synthetic-pretraining lane hinted at this direction but was walltime-limited.

One CTC decoder. All CER numbers use greedy best-path CTC decoding. Beam-search decoding with a language model would lower every reported CER but is unlikely to change the register-count *ordering*. The choice to skip it here is scope, not principle: comparing a language-model-augmented ViT-RGTS v2 against a language-model-augmented CNN–RNN would be a stronger absolute claim but would not affect the register-vs-CER question this report was set up to answer.

7 Conclusion and Future Work

This report set out to answer a question posed in the project’s opening emails: does the register-token intervention of Darcet *et al.*, originally proposed for classification Vision Transformers, transfer to a CTC-based Handwritten Text Recognition system, and does it produce attention maps that are “fine-grained and meaningful” enough to support downstream tasks such as writer identification?

The answer, after 63 training experiments across IAM and READ2016, is qualified. The recognition half of Darcet’s claim transfers cleanly: within a seed-noise floor of ± 0.08 pp, character error rate on IAM is flat across register counts $R \in \{0, 2, 4, 8, 16\}$, and the same near-flat pattern reproduces on READ2016 within 0.09 pp. Register tokens do not hurt recognition. The interpretability half transfers partially: register-added ViTs show measurably lower character-attention entropy, better spatial ordering, and more heads aligned to left-to-right reading order, but a residual sink pattern intrinsic to CTC self-attention survives and requires an additional query-side mechanism to fully suppress. The visualization style of Nikolaidou *et al.* [11] was reproduced end-to-end from our model: Fig. 3 shows the attention peak shifting from character to character along the word “booklet” using the same visualization recipe as the reference implementation. The best absolute recognizer on IAM is the compact CNN–RNN baseline with SWA at 4.62% CER; the best interpretable recognizer is ViT-RGTS v2 with $R = 4$ registers at 5.55% CER — a 0.93 pp CER cost for the interpretability the register mechanism was chosen to deliver.

Two ablations produce order-of-magnitude effects and are worth stating outside the main results: removing the CNN stem (or falling back to a row-major patch layout, as the legacy v1 architecture did) collapses CER from $\sim 6\%$ to $\sim 73\%$, isolating the strictly-left-to-right token-formation step as the single load-bearing design decision for a CTC ViT HTR encoder. This is the design constraint that separates a working from a non-working system, and it applies to any architectural variant of the same family (including HTR-VT-style plain-ViT encoders that use adaptive max-pooling in place of a CNN stem).

7.1 Immediate Extensions

Four extensions follow directly from the results of this report and are laid out in more detail in the Supplementary document.

Writer identification via VLAC over register-token features. The Souibgui *et al.* [13] pipeline aggregates character-indexed local descriptors into per-page writer representations. Feeding ViT-RGTS v2 attention-indexed patch features (at the CTC decoding column of each character) into a VLAC aggregator is a direct realization of the “downstream task” motivation from the problem statement, and the $R = 4$ model’s cleaner attention maps make it the natural feature extractor for this pipeline. This is the single most valuable follow-up.

CLS-augmented CTC ViT to close the residual sink gap. The residual sink pattern documented in Section 5.3 is mechanistically attributed to the absence of a CLS-style aggregate query. Adding a CLS token to ViT-RGTS v2 (queried by a small auxiliary head, *e.g.* line-level writer classification during training) should let the registers take over the global-information role more completely and close the token-norm asymmetry of Fig. 4(d). This is a small architectural change with a clear predicted signature.

Attention-entropy regularizer. CTC’s blindness to attention shape (Section 6.1) can be addressed by adding a low-weight entropy regularizer on the final-layer patch attention. This would provide the missing gradient signal that currently prevents registers from fully absorbing

the sinks. Care is needed: over-regularization would flatten attention rather than sharpen it, so the coefficient should be tuned against attention entropy on a held-out set, not against CER.

Synthetic→IAM curriculum. The synthetic pretraining lane in this report was walltime-truncated. A full curriculum — pretrain to convergence on the $\sim 950\text{K}$ synthetic corpus, fine-tune on IAM with a decayed learning rate — would test whether the CER gap between the from-scratch ViT-RGTS v2 (5.55%) and the CNN-RNN baseline (4.62%) can be closed by pretraining, without changing the register-attention story.

7.2 Closing Remark

The register-token mechanism does what its authors claim it does, subject to the structural differences between the setting it was proposed in and the setting it is imported into. On the accuracy side the transfer is clean. On the attention side the transfer is partial, and the mechanism of the partiality (patch tokens simultaneously acting as sink-carrying keys and CTC-decoding queries) is itself a research finding: it points at a specific class of extensions (CLS-augmentation, attention regularization) that would complete the transfer. The pipeline and the 63-run experiment matrix that produced these findings are released with the report so that subsequent work can build directly on top of them.

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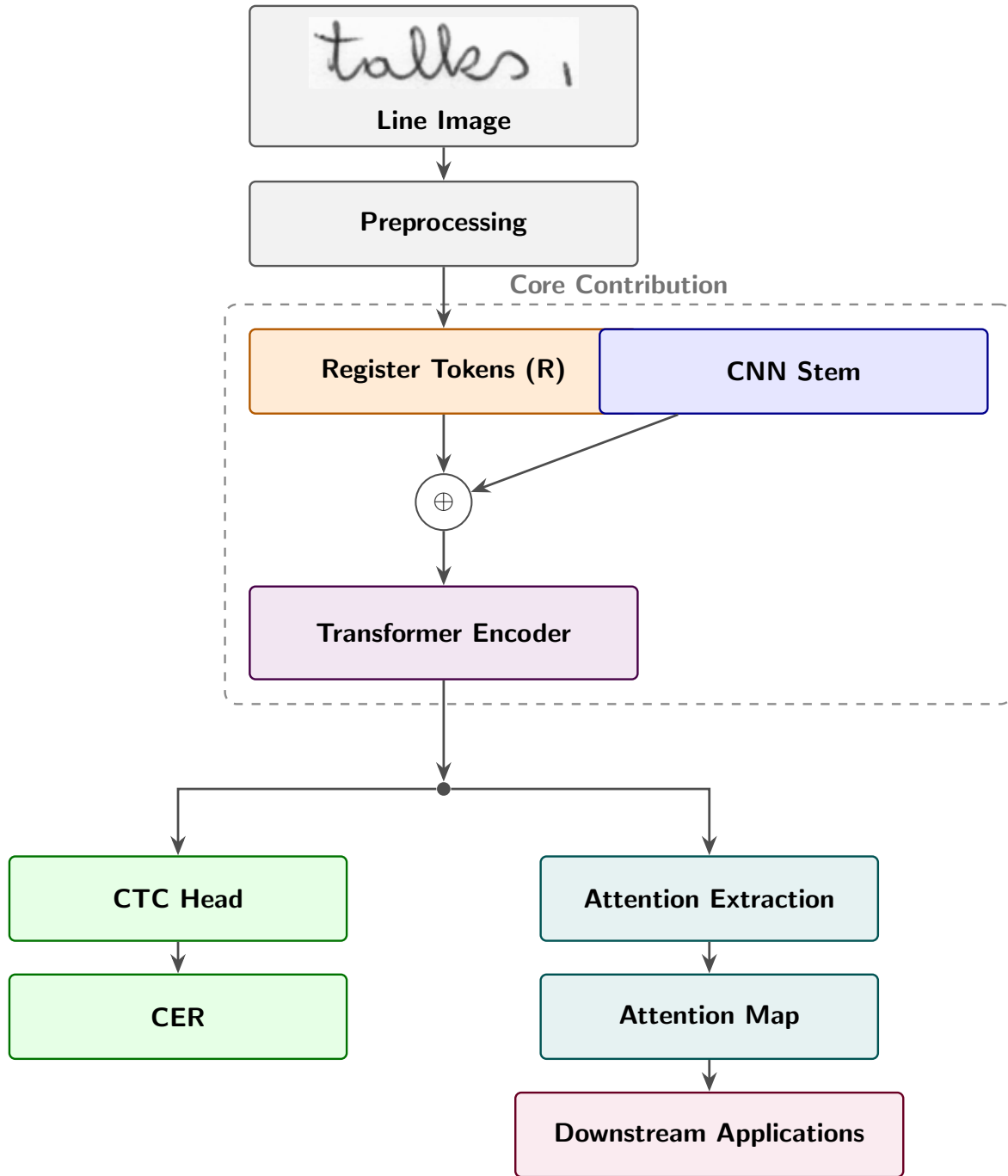


Figure 1: End-to-end HTR pipeline. A line image passes through preprocessing, one of four encoder families, and a CTC head; the same forward pass exposes attention tensors for interpretability analysis. The register tokens (green) are the additional learnable parameters this report studies. Downstream writer-identification is sketched but not implemented in the 10 ECTS project budget.

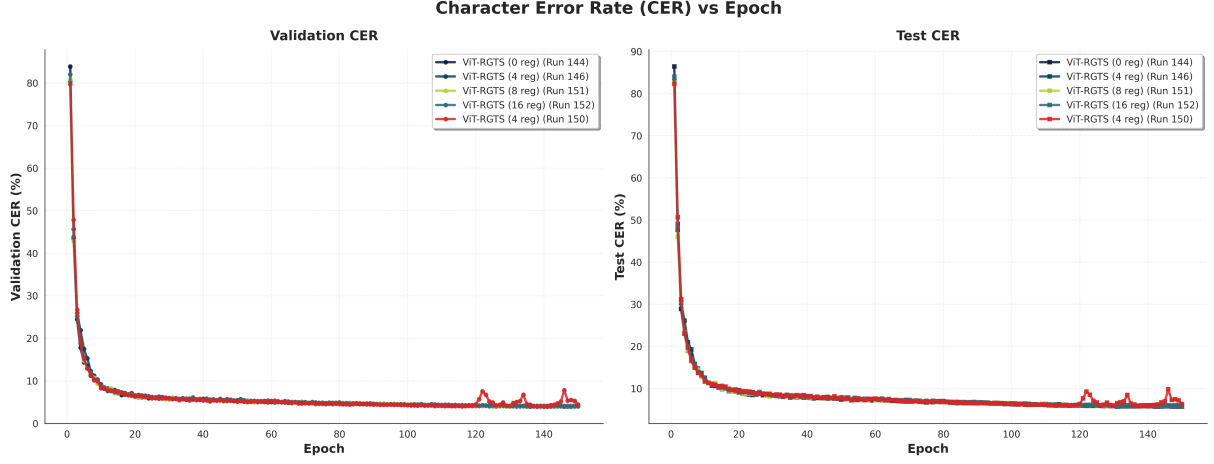


Figure 2: CER convergence of the ViT-RGTS v2 register sweep on IAM (best model `run_150` highlighted in red). All register counts follow the same trajectory after epoch 25; the CER-vs- R separation is smaller than the epoch-to-epoch fluctuation of any single run. Per-group ablation training curves are in the Supplementary document.



Figure 3: *Reproduction of Nikolaidou et al. Fig. 5 for our CTC ViT.* Top: input word crop (“booklet”, IAM a03-034-01-03). Rows 2–3: per-character attention overlay for the recognized characters b, o, k (row 2) and l, e, t (row 3), using the identical visualization recipe as *Beyond Memorization*. The attention peak shifts left-to-right along the word as the character index advances. Because our model is a CTC self-attention encoder and *Beyond Memorization* is a diffusion cross-attention decoder, the underlying attention mechanism is different, but the spatial shifting pattern that Nikolaidou *et al.* report is qualitatively reproduced.

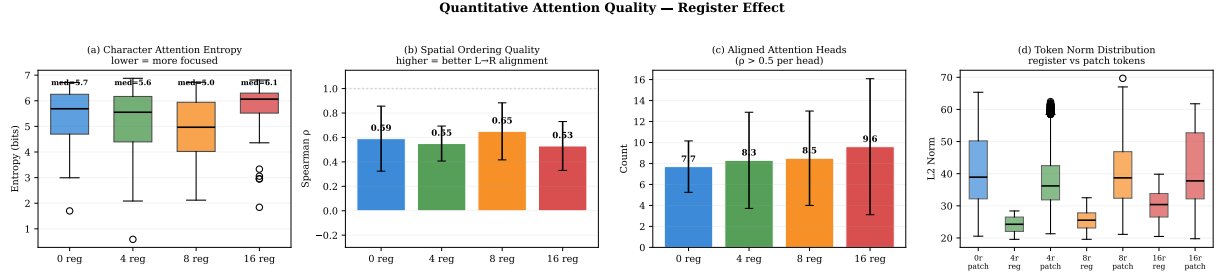


Figure 4: Attention-quality metrics as a function of register count on IAM. **(a)** Character attention entropy is lowest around $R \in \{4, 8\}$. **(b)** Spearman ordering correlation between character position and attention peak peaks at $R = 8$. **(c)** The number of self-attention heads aligned to left-to-right reading order (per-head $\rho > 0.5$) grows monotonically with R . **(d)** Token-norm distribution: register tokens (green, orange, red left bars) sit below patch tokens, showing that registers do carry a distinct role but do not fully absorb the high-norm sink pattern that would appear on patches in a classification ViT.